Same-depth inverse minima across a reset atom followed by 1113

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For a positive odd integer x, let

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

If $w = (a_1, \ldots, a_L)$ is an exact accelerated exponent word, put

$$B(w) := \sum_{i=1}^{L} a_i, \qquad C(w) := \sum_{i=1}^{L} 3^{L-i} 2^{a_1+\cdots+a_{i-1}}. \tag{2}$$

Thus

$$2^{B(w)} U_+^L(x) = 3^L x + C(w). \tag{3}$$

On an interior deficit-four record tail, write

$$A := 1, \qquad B := 12, \qquad C := 1113, \qquad D := 1122. \tag{4}$$

For a positive odd target y, define $\mathcal{R}_L(y; M)$ to be the set of reverse exponent strings $(d_1, \ldots, d_L)$ of total mass at most M which give a positive odd L-step predecessor of y. Explicitly, put

$$t_0 := y, \qquad t_i := \frac{2^{d_i} t_{i-1} - 1}{3}, \tag{5}$$

and require every t_i to be a positive odd integer. The forward word from t_L to y is $(d_L, \ldots, d_1)$.

Lemma 1 (inverse-mass recursion for the two reset–C blocks)**.** *Let $r \geq 0$.*

1. *If y is the endpoint of the exact word*

$$W_B(r) := 12\,1^r 1113, \tag{6}$$

then

$$\boxed{\mathcal{R}_{r+6}(y; r+9) = \{(3, 1^{r+3}, 2, 1)\}.} \tag{7}$$

2. If y is the endpoint of the exact word

$$W_D(r) := 1122\,1^r1113, \tag{8}$$

then

$$\boxed{\mathcal{R}_{r+8}(y; r+12) = \{(3, 1^{r+3}, 2, 2, 1, 1)\}.} \tag{9}$$

In each case the displayed reverse string is the reversal of the original word. Consequently the original word is the unique same-depth inverse word of minimum total exponent.

Proof. Let q be the start of the word. Direct composition of the four atom maps gives

$$\begin{aligned} 2^{r+9}y &= 3^{r+6}q + 13\,3^{r+4} - 2^{r+7} \quad \text{for } W_B(r), \\ 2^{r+12}y &= 3^{r+8}q + 137\,3^{r+4} - 2^{r+10} \quad \text{for } W_D(r). \end{aligned} \tag{10}$$

Hence the residue class of y modulo 3^L, where L is the relevant word length, is independent of q.

The reverse sets obey the finite recursion

$$\mathcal{R}_{\ell+1}(t; M) = \bigcup_{\substack{a\geq 1,\ a\leq M-\ell \\ 3\mid 2^a t-1}} \left\{ (a, \delta) : \delta \in \mathcal{R}_\ell\left(\frac{2^a t - 1}{3}; M - a\right) \right\}. \tag{11}$$

Only the residue of t modulo $3^{\ell+1}$ is used in this recursion. Substitution of the two residue classes from (10), followed by induction on r, gives the following complete list after the first $r+4$ reverse steps. The last two columns record the residue of the current state and the depth/mass still available.

family	live reverse prefix	mass	t_{r+4} mod 3	remaining depth/mass
$W_B(r)$	$1, 2, 1, 1^r, 3$	$r+7$	1	2/2
	$3, 1^{r+3}$	$r+6$	1	2/3
$W_D(r)$	$1, 2, 1, 1^r, 3$	$r+7$	1	4/5
	$3, 1^{r+3}$	$r+6$	1	4/6
	$3, 1^{r+2}, 3$	$r+8$	2	4/4.

(12)

The base case is obtained by four applications of (11). In the induction step, the only admissible transition which leaves enough mass for the remaining depth inserts one exponent 1 immediately before the final entry displayed in each prefix. This proves (12) for every r.

For completeness, the only lower-mass competitor through the terminal atom C begins with the reverse string

$$1, 2, 1, 1. \tag{13}$$

It is the half-seed branch. Pulling (13) through the following r reverse exponent-one steps reaches $(p-1)/2$. Both reset atoms satisfy

$$p + 1 \equiv 2 \pmod 3, \tag{14}$$

so $3 \mid (p-1)/2$ and this branch has no further inverse step.

The first row for $W_B(r)$ has current residue 1, so its next exponent is even and at least 2; no mass then remains for the final inverse step. The second row has exactly the mass needed for the forced suffix $(2, 1)$.

For $W_D(r)$, the first row takes a forced exponent 2 to another state of residue 1; a second forced exponent 2 then leaves only one unit of mass for two remaining steps. The third row takes a forced exponent 1 to a multiple of 3 and stops. The middle row has the unique forced suffix $(2, 2, 1, 1)$. This proves (7) and (9). □

Theorem 2 (reset–C same-depth inverse minimum)**.** *Let $r \geq 0$, let q be the start of one of the exact words $W_B(r)$ or $W_D(r)$, and put*

$$L := |W_B(r)| = r + 6 \quad \textit{or} \quad L := |W_D(r)| = r + 8. \tag{15}$$

If $y = U_+^L(q)$, then

$$\boxed{q = \min\{z \in 2\mathbb{Z} + 1 : z > 0,\ U_+^L(z) = y\}.} \tag{16}$$

The minimizer is unique.

Proof. Let $v = (c_1, \ldots, c_L)$ be any exact word from a positive odd z to y, and put $B' = B(v)$. Lemma 1 gives

$$B' \geq B_0, \qquad B_0 = \begin{cases} r + 9, & W_B(r), \\ r + 12, & W_D(r). \end{cases} \tag{17}$$

For $W_B(r)$, the start of the record atom B is congruent to 11 modulo 16. The value $q = 11$ would make its endpoint 13, whose next exponent is 3, whereas the next atom in $W_B(r)$ starts with exponent 1. Hence $q \geq 27$. For $W_D(r)$, the exact D congruence is $q \equiv 71 \pmod{128}$, so $q \geq 71$. In particular,

$$q > 2^{B_0+1-L} = \begin{cases} 16, & W_B(r), \\ 32, & W_D(r), \end{cases} \tag{18}$$

and the original affine equation implies $y > (3/2)^L$.

If $B' = B_0$, the same lemma says that v is the original word, so $z = q$.

Assume that $B' \geq B_0 + 1$. For the partial sums in (2), positivity of the exponents gives

$$c_1 + \cdots + c_{i-1} \leq B' - (L - i + 1).$$

Consequently

$$\begin{aligned} C(v) &\leq 2^{B'-L} \sum_{i=1}^{L} 3^{L-i} 2^{i-1} \\ &= 2^{B'-L}(3^L - 2^L) < 2^{B'-L} 3^L. \end{aligned} \tag{19}$$

Equations (3) and (19), together with $y > (3/2)^L$, imply

$$z > \frac{2^{B'} y}{3^L} - 2^{B'-L} \geq \frac{2^{B_0+1} y}{3^L} - 2^{B_0+1-L}. \tag{20}$$

If C_0 is the carry of the original word, then $2^{B_0} y = 3^L q + C_0$, and hence

$$z > 2q + \frac{2C_0}{3^L} - 2^{B_0+1-L}. \tag{21}$$

Since $C_0 > 0$, equation (18) makes the right side of (21) strictly larger than q. This proves (16) and uniqueness. □

Corollary 3 (one-reset cross-block descent is impossible)**.** *Suppose that an interior record tail contains a block BA^rC or DA^rC. No positive odd inverse branch of the same raw depth reaches the terminal endpoint from a state smaller than the block's actual start. Therefore a descent argument which crosses a reset blocker before a C half-coalescence must either change the inverse depth or span at least two nontrivial record atoms.*

Proof. The first assertion is Theorem 2. An inverse word spanning precisely the displayed block has the same raw depth, so the second assertion follows. □